REVIEW

Pharmacology of thoracic anesthesia

Tamás Végh[a,b], Katalin Szamos[a] and Boglárka Balla[a]

Purpose of review
Thoracic anesthesia is one of the most demanding subspecialties, requiring the integration of surgical conditions with patient safety. Procedures such as lung resections, video-assisted thoracic surgery, robotic surgery, and transplantation impose unique physiologic stress, particularly during one-lung ventilation, which challenges gas exchange and cardiovascular stability. This review summarizes current evidence on anesthetic strategies in thoracic surgery, including induction and maintenance techniques, opioid use, neuromuscular blockade, and hemodynamic management.

Recent findings
Thoracic anesthesia present ongoing debates. No consensus exists on whether total intravenous anesthesia or volatile anesthesia is superior: propofol may provide long-term oncologic benefit, whereas volatiles offer pulmonary protection but raise concerns about tumor progression and impaired hypoxic pulmonary vasoconstriction. Opioids remain effective but carry risks of respiratory depression, delirium, hyperalgesia, immune suppression, and persistent use, fueling opioid-sparing but not opioid-free strategies. Neuromuscular blockade improves surgical conditions, yet optimal depth outside robotic surgery and the need for routine reversal remain debated. Hemodynamic management balances euvolemia against hypovolemia.

Summary
Modern thoracic anesthesia pharmacology combines novel agents with established drug classes, integrating current evidence to improve intraoperative stability, reduce complications, and promote recovery.

Keywords
anesthetics, muscle relaxants, opioids, thoracic anesthesia

INTRODUCTION

Thoracic anesthesia is one of the most demanding areas of anesthetic practice, requiring precise integration of surgical needs with patient safety. Procedures such as lung resections, video-assisted thoracic surgery (VATS), robotic surgery, and transplantation impose unique physiological stress, especially during one-lung ventilation (OLV), which challenges both gas exchange and cardiovascular stability [1,2■■]. Management must therefore maintain oxygenation, ensure hemodynamic balance, limit postoperative pulmonary complications, and support recovery [3■■].

In recent decades, practice has shifted from reliance on opioids and volatile agents toward individualized, multimodal, and opioid-sparing strategies, reflecting awareness of long-term consequences on inflammation, immunity, and oncologic outcomes [4]. Propofol-based total intravenous anesthesia (TIVA) and volatile anesthesia both offer advantages, with no universal superiority. Similarly, approaches to neuromuscular blockade, hemodynamics, infection prophylaxis, and blood management emphasize patient-specific tailoring guided by quantitative monitoring [5–7].

Modern thoracic anesthesia is thus best viewed as a multidisciplinary, evidence-based framework, integrating perioperative strategies to optimize safety and outcomes.

INDUCTION OF ANESTHESIA

Induction for thoracic surgery must be individualized based on physiology, comorbidities, airway anatomy, surgical plan, and institutional expertise, as no single method is universally superior. Intravenous induction provides a rapid onset, predictable depth, tight hemodynamic control, less postoperative nausea and vomiting (PONV), and favorable recovery, making propofol-based TIVA

[a]Department of Anesthesiology and Intensive Care, University of Debrecen, Debrecen, Hungary and [b]Outcomes Research Consortium, Houston, Texas, USA

Correspondence to Tamás Végh, MD, PhD, Department of Anesthesiology and Intensive Care, University of Debrecen, Nagyerdei krt. 98, Debrecen 4032, Hungary
Tel: +36 52 255 347; e-mail: veghdr@med.unideb.hu

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KEY POINTS

- Despite differing anti-inflammatory profiles between volatile and intravenous agents, current evidence shows no significant advantage of one technique over the other in terms of postoperative or oncologic outcomes.
- Modern thoracic anesthesia favors multimodal, opioid-sparing strategies to reduce side effects and potential immunosuppressive or oncologic risks while ensuring adequate analgesia.
- Deep neuromuscular block improves surgical conditions in video-assisted thoracic surgery and robotic surgery but requires continuous quantitative monitoring and appropriate reversal to prevent residual paralysis.
- Intraoperative stability should be maintained through physiology-guided fluid and vasoactive therapy, aiming for euvolemia and tailored cardiovascular support based on individual patient profiles.

ideal in elderly or cardiovascularly fragile patients [8▪▪,9]. Inhalational induction remains valuable when maintaining spontaneous ventilation is critical, such as in pediatrics, difficult airways, mediastinal mass, or airway trauma; sevoflurane offers acceptable intubating conditions and supports fiberoptic strategies. Airway considerations often dictate the choice: intravenous induction offers a reliable hypnotic 'safety net', while inhalational induction allows incremental depth with spontaneous breathing [10–12]. Each technique has pitfalls – intravenous agents risk hypotension or respiratory depression, while inhalational methods may cause excitatory responses. Best practice integrates patient-specific factors with surgical needs, using both methods complementarily within a collaborative, patient-centered approach [13].

MAINTENANCE OF ANESTHESIA

The influence of anesthetic technique on clinical outcomes in thoracic surgery remains a subject of debate. Available studies differ substantially in methodology and have reported inconsistent findings. Some evidence indicates that patients exposed to extensive surgical trauma, such as pneumonectomy, might gain the most from the anti-inflammatory actions of volatile anesthetics [14▪▪,15], although further trials are required to confirm this observation.

Comparisons between volatile halogenated agents and intravenous propofol have repeatedly highlighted differences in inflammatory responses. Most investigations suggest that inhalational agents are associated with lower levels of alveolar and, potentially, systemic inflammation [16–19]. Additional experimental and clinical data further indicate that desflurane and sevoflurane mitigate the inflammatory activation of epithelial cells and dampen mediator release in bronchoalveolar lavage fluid, possibly by preserving the integrity of the endothelial glycocalyx [16–22].

When postoperative outcomes are examined, however, the literature is less consistent. Some studies reported that volatile anesthesia was linked to fewer pulmonary complications compared with propofol during OLV [16,17,23]. A meta-analysis indicated that the use of inhalational anesthetics was associated with a marked reduction in postoperative pulmonary complications and a shorter duration of hospitalization [20]. By contrast, findings from a large multicenter randomized trial involving thoracic surgery patients did not reveal any meaningful difference in complication rates between desflurane and propofol. Subgroup analyses from that study suggested, however, that patients exposed to the most extensive surgical trauma might still gain benefit from the anti-inflammatory properties of volatile agents [15].

Historically, enthusiasm for propofol-based TIVA stemmed from laboratory studies demonstrating suppression of tumor-promoting pathways, inhibition of angiogenesis, and preservation of natural killer (NK) cell activity. Conversely, volatile anesthetics were shown in preclinical models to promote angiogenesis, epithelial–mesenchymal transition, and metastasis while impairing NK-cell function; however, these mechanistic data – derived largely from reductionist in-vitro systems and rodent models – did not translate into clinical benefit [24,25▪]. Large randomized trials have consistently failed to show oncologic superiority of one maintenance technique over the other. The BCA-R trial randomized more than 2100 [26] women undergoing curative breast cancer surgery to propofol with paravertebral block versus sevoflurane with opioid analgesia, and after 3 years' follow-up there was no difference in recurrence-free survival or persistent pain. Similarly, Cao *et al*. [27] reported on nearly 1200 elderly patients undergoing major cancer surgery in China and found no significant differences in overall survival, recurrence-free survival, or event-free survival between propofol–TIVA and sevoflurane anesthesia. A recent review explained why strong mechanistic signals failed to materialize clinically: NK-cell assays and invasion models overestimated relevant effects, rodent metastasis models bypassed early metastatic steps, and many preclinical studies lacked methodological rigor [28▪]. Across three large randomized clinical trials in breast, lung, and abdominal cancers ($N \approx 4770$), no benefit of regional or propofol-based anesthesia on cancer recurrence was demonstrated [15,29,30]. The conclusion is that anesthetic technique itself does not influence long-term oncologic outcomes.

Thoracic oncology has received particular attention because both non-small-cell lung cancer (NSCLC) and esophageal cancer resections are highly morbid procedures in which recurrence strongly influences survival. In NSCLC, the international GAS-TIVA trial is the first large randomized controlled trial designed to test whether propofol–TIVA improves recurrence-free survival after curative resection compared with volatile anesthesia; results are pending [31].

Retrospective studies remain inconsistent. Zhu *et al.* [32] found that opioid and glucocorticoid exposure predicted prognosis more strongly than anesthetic type, with higher fentanyl doses paradoxically associated with improved overall survival but also more pulmonary complications. Seo *et al.* [33] analyzed more than 1500 stage I–II patients and suggested longer recurrence-free and overall survival with TIVA, while Hayasaka *et al.* [34] also reported a prognostic advantage for propofol. The lack of benefit from propofol-based TIVA has also been described for lung surgeries [35,36]. In esophageal cancer, a randomized trial of 553 patients, volatile versus TIVA during minimally invasive esophagectomy was compared and showed fewer pulmonary complications with volatile anesthesia (36.5 versus 47.5%), though no oncologic differences were observed [37▪▪]. Retrospective analyses also found no immunoinflammatory advantage for either approach [38].

Earlier, it has been demonstrated that adding epidural analgesia to general anesthesia in lung cancer surgery did not improve recurrence-free survival [39▪]. A subgroup analysis found more favorable immune profiles with epidural analgesia compared with oral opioids, again suggesting that analgesic management may influence perioperative immunity more than anesthetic maintenance [40]. Watanabe *et al.* [41] argued that anesthetic and analgesic strategies may shift the timing, but not the incidence, of recurrence. Collectively, the evidence indicates that anesthetic maintenance does not alter long-term oncologic outcomes, while perioperative drug exposures, analgesic strategies, and pulmonary complications are more decisive. A retrospective clinical study [28▪] revealed that in patients with advanced lung cancer treated with immune checkpoint inhibitors, the use of opioids may diminish the therapeutic efficacy of these inhibitors, resulting in significantly poorer clinical outcomes [42].

An important short-term difference between techniques lies in oxygenation during OLV. Hypoxic pulmonary vasoconstriction (HPV) is a protective mechanism that reduces shunt and maintains oxygenation, but volatile anesthetics impair HPV in a dose-dependent fashion, particularly above 1.5 minimum alveolar concentration (MAC), through K^+-channel activation, nitric oxide signaling, and altered calcium influx [43]. Clinically, this can manifest as early intraoperative hypoxemia, more frequent with volatiles than with propofol–TIVA [44]. Under TIVA, HPV is better preserved, and patients often exhibit higher PaO_2 during the first 15–30 min of OLV. Although these episodes are usually transient, they may be clinically important in patients with pulmonary hypertension, emphysema, or impaired vascular reactivity [45,46]. Management includes lowering volatile concentration, using balanced anesthesia, applying Positive end-expiratory pressure (PEEP) and recruitment maneuvers, or providing Continuous positive airway pressure (CPAP) or oxygen to the operative lung when feasible [47,48,49▪].

OPIOID USE IN THORACIC ANESTHESIA

The role of opioids in perioperative analgesia for thoracic noncardiac surgery has undergone substantial change over time. Once central to pain control, they are now integrated into multimodal strategies within enhanced recovery after thoracic surgery pathways, where their use is generally limited to rescue therapy [50]. While opioids remain highly effective for nociceptive suppression, their side effects – such as respiratory depression, nausea, ileus, opioid-induced hyperalgesia, and potential immunosuppressive effects via NK-cell suppression – have driven the adoption of opioid-sparing approaches [51–53]. Regional techniques, including thoracic epidural and paravertebral blocks, along with adjuncts such as ketamine, dexmedetomidine, lidocaine, magnesium, nonsteroidal anti-inflammatory drugs, and acetaminophen, are increasingly favored to reduce systemic opioid exposure, though complete opioid avoidance is rarely feasible given the severity of pain after thoracic procedures, particularly open resections. Remifentanil is often preferred intraoperatively for its rapid titratability and favorable recovery profile, and current best practice emphasizes judicious, opioid-sparing rather than opioid-free anesthesia [54,55▪].

In the oncologic context, growing interest surrounds the potential influence of opioids on long-term outcomes such as recurrence, metastasis, and survival, although evidence remains inconsistent across lung cancer and other malignancies. A new research paradigm, 'precision oncoanalgesia', seeks to clarify these relationships by integrating tumor-specific genomic and transcriptomic data into perioperative planning, paralleling the principles of precision oncology. This approach aims to tailor analgesic regimens to the molecular characteristics of each patient's tumor, potentially identifying those who may benefit from opioid-sparing strategies and those for whom opioids can be used without adverse oncologic consequences [56▪▪].

Early studies suggest that tumor genomics can modify the relationship between analgesic dose and cancer outcomes, shifting the effect in either a protumor or antitumor direction [57▪]. Du *et al.* [58] reported worse oncological outcomes with higher opioid exposure in esophagectomy, while other investigations indicate that analgesic technique may shape perioperative immune responses and recurrence dynamics. Looking forward, precision oncoanalgesia may provide a foundation for personalized analgesic plans that not only improve perioperative pain management but also optimize long-term oncologic results.

NEUROMUSCULAR BLOCKADE

Neuromuscular blockade strategies should be individualized and grounded in quantitative monitoring [59].

Steroid neuromuscular blocking agents (NMBAs) such as rocuronium are recommended (atracurium remains an alternative in severe hepatic/renal impairment) because the availability of sugammadex allows rapid and complete reversal – even from deep block – shortening time to extubation and avoiding cholinergic adverse effects associated with acetylcholinesterase inhibitors (e.g. bradycardia, increased airway secretions, and paradoxical weakness) [60]. Choosing the NMBA depends on patient comorbidities, expected duration, and the team's experience, but deep block clearly improves surgical conditions for VATS and robotic procedures by minimizing diaphragmatic motion and cough; that benefit must be balanced against the risk of residual paralysis and postoperative pulmonary complications if monitoring and reversal are inadequate [61]. Continuous infusions of NMBAs do not necessarily improve outcomes beyond what can be achieved with carefully titrated boluses, and the overall evidence regarding deep versus moderate block outside robotic cases remains mixed. The safest approach is to consider deep block when it confers surgical advantage – particularly in robotic thoracic surgery – while ensuring continuous quantitative neuromuscular monitoring throughout and using appropriate reversal before extubation [62▪].

Nonrelaxant techniques remain appropriate in selected populations, notably patients with mediastinal mass syndrome or myasthenia gravis, in whom maintaining spontaneous ventilation or minimizing NMBA exposure is desirable [63–66].

Routine reversal in all cases is not recommended; excellent analgesia and stable anesthesia do not always require repeated NMBA dosing, and reversal should be guided by quantitative data.

Monitoring must begin before the first NMBA dose and continue until full recovery, with a target train of four (TOF) ratio greater than 0.9 before extubation. Sugammadex is the reversal agent of choice when needed; routine neostigmine is discouraged in thoracic patients because it increases airway secretions and can provoke bradycardia, while the anticholinergics coadministered to counter these effects may themselves cause tachycardia and potentially worsen postoperative cardiovascular risk [67–69].

HEMODYNAMIC MANAGEMENT

Hemodynamic instability is common in thoracic surgery because of OLV, surgical manipulation, and comorbid cardiopulmonary disease, so management must be physiology-driven. The choice of inotropes or vasopressors depends on baseline ventricular function and the intraoperative inflammatory state. Transesophageal echocardiography is recommended in high-risk patients to guide fluid therapy and assess ventricular performance. Excess fluid administration impairs postoperative pulmonary function, while hypovolemia causes metabolic and hemodynamic compromise; the target is euvolemia [70–72,73▪].

In reduced ventricular function, agents such as dobutamine can support cardiac output, and in severe dysfunction, preoperative levosimendan may be considered. In low systemic vascular resistance states related to inflammation, norepinephrine or vasopressin can help restore perfusion pressure once adequate preload is ensured. If both ventricular performance and systemic resistance are impaired, combined inotropes and vasopressors under echocardiographic guidance are indicated. Severe right ventricular dysfunction requires careful pressure maintenance, strict avoidance of fluid overload, reduction of pulmonary vascular resistance with agents such as inhaled nitric oxide or sildenafil, and avoidance of hypercapnia [7,74]

Desaturation during OLV is usually managed with conventional measures, though pharmacologic adjuncts may be needed in refractory hypoxemia. Inhaled nitric oxide or aerosolized prostacyclin selectively dilates vessels in ventilated lung regions and may be combined with systemic vasoconstrictors or pulmonary vasodilators in decompensated pulmonary hypertension with right heart failure. Almitrine can augment HPV but is rarely indicated, especially with volatile anesthesia. Milrinone increases contractility and reduces pulmonary pressures, while sildenafil lowers right ventricular afterload and controls pulmonary hypertension in selected cases. If desaturation is linked to hypotension, vasopressors such as phenylephrine, ephedrine, norepinephrine, or vasopressin restore systemic pressure without substantially worsening pulmonary vascular resistance [75–79].

CONCLUSION

Thoracic anesthesia demands a nuanced, physiology-driven approach that balances anesthetic technique, analgesia, neuromuscular blockade, hemodynamics, and blood management. Evidence does not establish universal superiority of intravenous or inhalational strategies; rather, choice should be individualized. Opioid-sparing multimodal analgesia, quantitative NMBA monitoring with sugammadex reversal, and vigilant hemodynamic optimization are cornerstones of perioperative care. Restrictive, goal-directed transfusion with point-of-care guidance minimizes complications, while cell salvage and antifibrinolytics provide valuable adjuncts in selected settings. Ultimately, modern thoracic anesthesia prioritizes safety, pulmonary protection, and recovery, guided by evidence-based protocols, multidisciplinary collaboration, and individualized decision-making tailored to patient comorbidities and surgical context.

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Conflicts of interest

There are no conflicts of interest.

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■ special interest.

■■ outstanding interest.

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The review outlines that in thoracic anesthesia there's a very narrow 'safe zone' between fluid excess (which risks pulmonary edema) and insufficient volume (which risks hypoperfusion or renal injury), and while liberal fluid strategies are rejected, evidence remains scarce for optimal restrictive or goal-directed regimens, with dynamic monitoring and balanced crystalloids favored.

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